

How do you know if a product is doing well?

Success Metric

An algorithm measuring performance

How do companies (often) measure success?

- Pageviews
- Clicks
- Uptime
- Latency
- Active Users
- Revenue

But the UX side is missing !!!

UX Success Metric

A user-centered algorithm measuring performance

Goal of UX Metric

Implement and track user-centered metrics for the product in service of company goals

Rational of UX Metric

Measuring UX with products is necessary to shape future direction and understand successes and failures.

Challenge

Metrics are imperfect and when poorly chosen or optimized without regular validation will lead to product decisions in the wrong direction.

Is this a happy story?



Maybe not

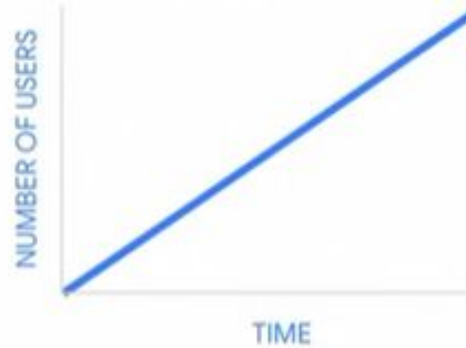
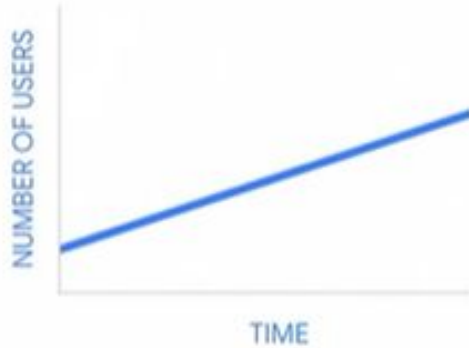
30-Day Actives

=

New Users

+

Existing Users



Who is in charge to create UX metrics?

UX researchers shape user-centered metrics

How do we create UX metrics?

Key concepts to help you define metrics:

Goal: How do you want users to engage with the product?



Signal: What behaviors or attitudes would indicate your goal has been met or failed? Mapping your goals to those actions can let you know whether you're on track.



Metric: Distill signals into trackable metrics that you can monitor in a real-time dashboard.

Example: Searching products in an e-commerce site

Goal: Users find what they need on the first try.

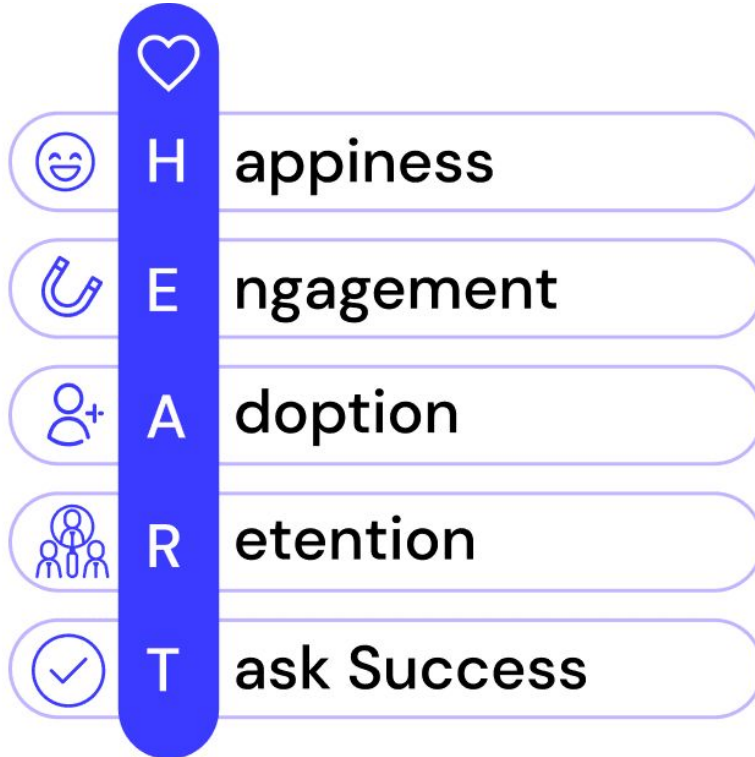
Signal: Length of time before coming back to search again after clicking on search result.

Metric: Average length of time before coming back to search after clicking on search result.

What is the Google HEART Framework?

The HEART framework is a set of user-centered metrics. It was developed to evaluate the quality of the user experience, and help teams measure the impact of UX changes.

Measuring UX with HEART



Adoption:

Goal: Users discover and try the product/service.

Signal: Signing up for an account.

Metric: Registration rate.

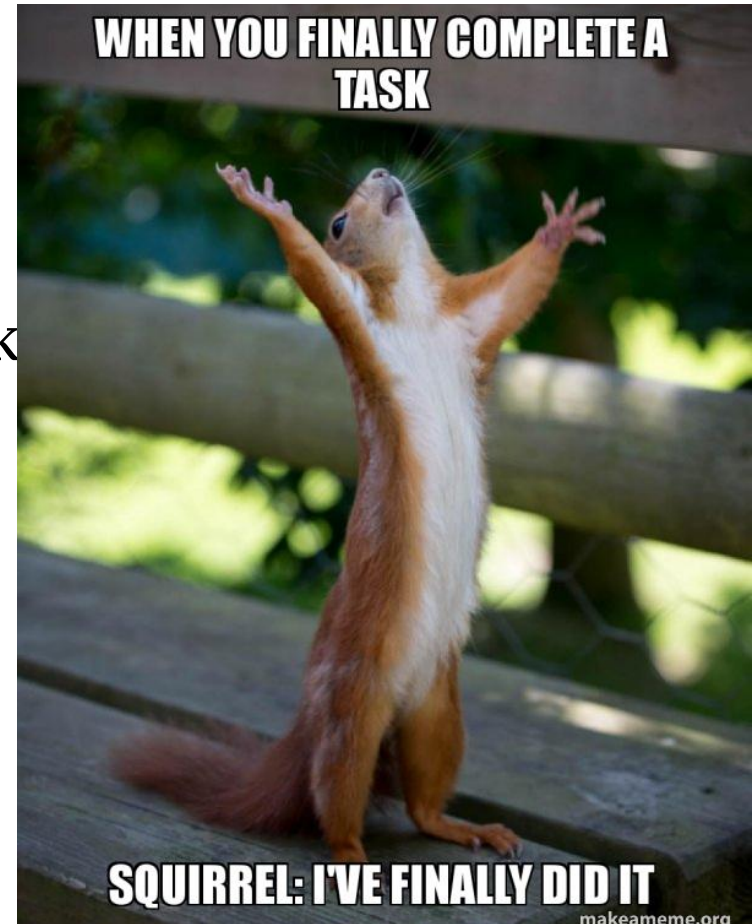


Task success:

Goal: Users can complete the task effortlessly

Signal: Completing task efficiently.

Metric: Error rate vs completion rate.

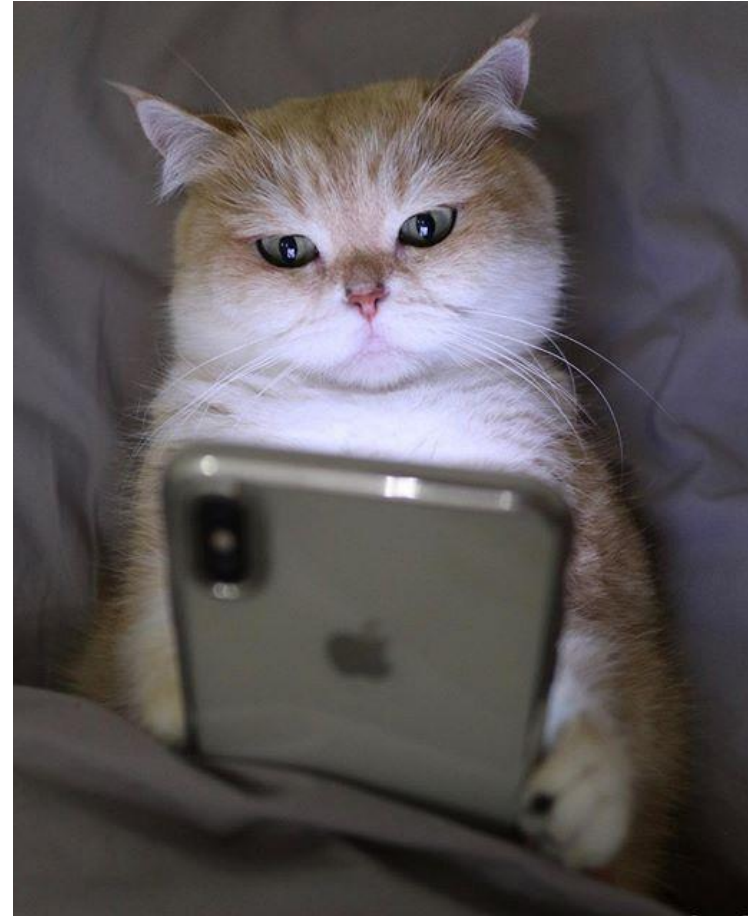


Engagement:

Goal: Users interact with the product/service deeply and broadly.

Signal: Frequently using the app.

Metric: Number of visits per user per week, session length.



Retention:

Goal: Users keep coming back to the product/service.

Signal: Renewing a subscription.

Metric: Subscription renewal rate.



Happiness

Goal: Users enjoy/delight in using the product/service.

Signal: 5 star rating.

Metric: Net promoter score



	<i>Goals</i>	<i>Signals</i>	<i>Metrics</i>
<i>Happiness</i>			
<i>Engagement</i>			
<i>Adoption</i>			
<i>Retention</i>			
<i>Task Success</i>			

	Goals	Signals	Metrics
Happiness	Users find the app helpful, fun, and easy to use	• Responding to surveys • Leaving 5-star ratings • Leaving user feedback	• Net Promoter Score • Customer satisfaction rating • Number of 5-star reviews
Engagement	Users enjoy the app content and keep engaging with it	• Spending more time in the app	• Average session length • Average session frequency • Number of conversions (consuming content, uploading files, purchases, etc.)
Adoption	New users see the value in the product or new feature	• Downloading, launching app • Signing up for an account • Using a new feature	• Download rate • Registration rate • Feature adoption rate
Retention	Users keep coming back to the app to complete a key action	• Staying active in the app • Renewing a subscription • Making repeat purchases	• Churn rate • Subscription renewal rate
Task Success	Users complete their goal quickly and easily	• Finding and viewing content quickly • Completing tasks efficiently	• Search exit rate • Crash rate

Triangulate metrics

Triangulating metrics to improve decisions



Important

- Choose **signals** that **indicates goal success**, not just those that are readily available.
- You don't need metrics for every category - start with the minimum set and go from there.
- You don't always need multiple metrics per category.
- Incorporate UX metrics reviews into business reviews.

Person's law: “That which is measured improves. That which is measured and reported improves exponentially”

Goodhart's law: “When a measure becomes a target, it ceases to be a good measure”

Revisit the metrics frequently, adapt and iterate.

When we should use HEART Framework?

(B2C) only for consumer products, where the user can decide whether to use the product.

Then what about (B2B)?

Let's say, products that people use for their work (such as enterprise products, intranets, healthcare systems, government tools, and many other types of complex apps), where the end user may not get to choose whether to use your product.

The CASTLE Framework

CASTLE

- C = Cognitive load
- A = Advanced feature usage
- S = Satisfaction
- T = Task efficiency
- L = Learnability
- E = Errors

Cognitive Load

The level of **mental effort** required to complete key tasks. It includes the user's **working-memory burden** while they complete workflows, as well as how easily an interrupted task or workflow can be resumed.

Tool: **NASA-TLX**, **PURE** or **cognitive walkthrough**

Example

	Goal	Signal	Measures
Cognitive Load	Notifications are not too frequent or intrusive.	Users self-reported level of mental workload decreases.	NASA-TLX

Advanced Feature Usage

The **usage** rates of **optional features** — what proportion of users interact with features that they technically are not required to use but would be beneficial to them. It helps the team **evaluate** the **discoverability**, **clarity**, or **usefulness** of these features, and to help prioritize future feature roadmaps.

Example

	Goal	Signal	Measures
Advanced features	Data dashboard presents information that is helpful to individual users.	Users customize their dashboard widgets to personal preferences.	Rate of accounts that customize dashboard widgets, most common customization choices (tracked through analytics)

Satisfaction

How the user **feels** about using the product. Analytics will not be an appropriate way to measure satisfaction, since it is not directly observable. We recommend a mix of **surveys**, post-task **questionnaires** in **quantitative usability research**, and analytics signals such as “**rage clicks**”

Example

	Goal	Signal	Measure
Satisfaction	Users enjoy working with the product	Self-reported satisfaction	Questionnaire: SEQ

Task Efficiency

It measures how quickly users can successfully complete key workflows. Efficiency can be quantified as time on task or the number of steps that a user must undertake to complete their workflow.

Example

	Goals	Signals	Measures
Efficiency	Users can complete key tasks without getting sidetracked.	Users complete workflows quickly.	Time on task (measured in quantitative usability testing)
		Users complete workflows with the fewest number of steps necessary.	Deviations from “optimal” task procedure (obtained through log analysis)

Learnability

It covers the effort, time, and resources needed for a new user to onboard and learn how to use the product effectively in their role..

Example

	Goal	Signal	Measure
Learnability	Users can recall how to complete a key workflow.	User task performance improves over time.	Time on task (measured longitudinally, in a quantitative usability test with within-subjects design)

Errors

It covers both **data-quality issues** (such as accidental data-entry slips or misunderstanding-driven mistakes), as well as **system-error messages** that **stop users** in their tracks. Instrumenting error messages, exploring more nuanced **indicators** of **errors**, such as repetitive behaviors that might indicate that the user is experiencing frustrating loops.

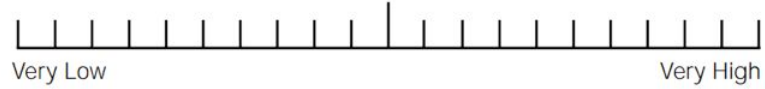
Example

	Goals	Signals	Measures
Errors	User-entered data improves in accuracy.	Users don't input values that are outside of an acceptable range.	Validation errors (tracked through analytics)

NASA-TLX: Post-Task Workload

Mental Demand

How mentally demanding was the task?



Physical Demand

How physically demanding was the task?



Temporal Demand

How hurried or rushed was the pace of the task?



Performance

How successful were you in accomplishing what you were asked to do?



Effort

How hard did you have to work to accomplish your level of performance?

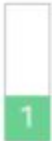


Frustration

How insecure, discouraged, irritated, stressed, and annoyed were you?



PURE Scores: Measures of Friction

	The step can be accomplished easily by the target user, due to low cognitive load or because it's a known pattern, such as the acceptance of a terms-of-service agreement.
	The step requires a notable degree of cognitive load (or physical effort) by the target user, but can generally be accomplished with some effort.
	The step is difficult for the target user , due to significant cognitive load or confusion; some target users would likely fail or abandon the task at this point.

PURE score for a given task is simply the sum of the scores of all steps' ratings in that task.

